

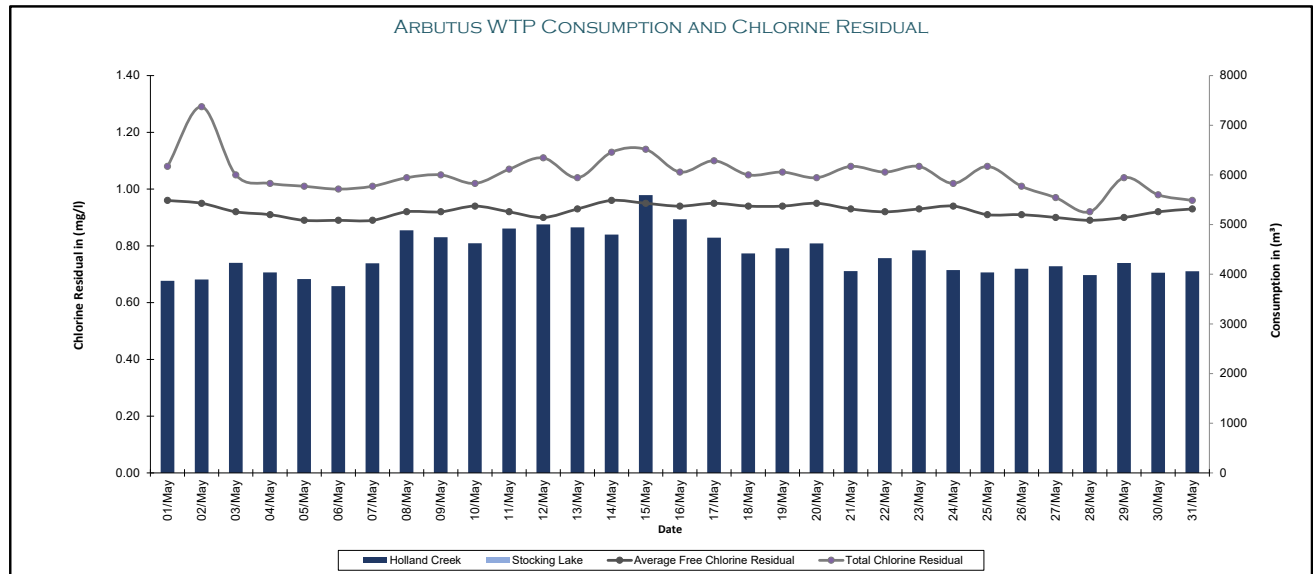
TOWN OF LADYSMITH - ARBUTUS WATER TREATMENT PLANT

MAY 2024 - MONTHLY REPORT

Date	Daily Flow			Chlorine Residual				CT*	External Lab Testing					
	Stocking Lake	Holland Creek	Combined Flow	Free Min	Free Max	Free Avg	Total		HPC	E.coli	Total Coliforms	Aluminum	THM	HAA
	m³	m³	m³	mg/l	mg/l	mg/l	mg/l		CFU	MPN	MPN	mg/l	mg/l	mg/l
01-May	0	3869	3869	0.96	0.97	0.96	1.08	128						
02-May	0	3894	3894	0.95	0.98	0.95	1.29	222	1	< 1	< 1			
03-May	0	4231	4231	0.92	0.95	0.92	1.05	166						
04-May	0	4037	4037	0.89	0.92	0.91	1.02	149						
05-May	0	3904	3904	0.89	0.91	0.89	1.01	158						
06-May	0	3761	3761	0.88	0.90	0.89	1.00	180						
07-May	0	4221	4221	0.87	0.90	0.89	1.01	177	< 1	< 1	< 1			
08-May	0	4885	4885	0.88	0.92	0.92	1.04	156						
09-May	0	4744	4744	0.91	0.93	0.92	1.05	222						
10-May	0	4624	4624	0.91	0.94	0.94	1.02	212						
11-May	0	4920	4920	0.92	0.94	0.92	1.07	160						
12-May	0	5002	5002	0.89	1.11	0.90	1.11	184						
13-May	0	4945	4945	0.87	0.95	0.93	1.04	270						
14-May	0	4799	4799	0.92	0.96	0.96	1.13	213						
15-May	0	5594	5594	0.94	0.97	0.95	1.14	192	< 1	< 1	< 1	0.0124	0.0129	0.0083
16-May	0	5108	5108	0.92	0.95	0.94	1.06	264						
17-May	0	4736	4736	0.93	0.95	0.95	1.10	228						
18-May	0	4419	4419	0.94	0.95	0.94	1.05	155						
19-May	0	4523	4523	0.93	0.95	0.94	1.06	149						
20-May	0	4620	4620	0.93	0.95	0.95	1.04	89						
21-May	0	4064	4064	0.93	0.95	0.93	1.08	146						
22-May	0	4323	4323	0.91	0.93	0.92	1.06	157	< 1	< 1	< 1			
23-May	0	4482	4482	0.90	0.93	0.93	1.08	139						
24-May	0	4083	4083	0.95	0.92	0.94	1.02	121						
25-May	0	4036	4036	0.90	0.94	0.91	1.08	158						
26-May	0	4110	4110	0.89	0.91	0.91	1.01	154						
27-May	0	4160	4160	0.89	0.91	0.90	0.97	117						
28-May	0	3983	3983	0.89	0.90	0.89	0.92	160						
29-May	0	4226	4226	0.88	0.90	0.90	1.04	164						
30-May	0	4032	4032	0.89	0.92	0.92	0.98	160	< 1	< 1	< 1			
31-May	0	4060	4060	0.91	0.94	0.93	0.96	127						

*CT - Recorded as the minimum value at the highest daily flow ** Manual Residual were not taken

Total	0	136395	136395											
Average	0	4400	4400	0.91	0.94	0.92	1.05	170	< 1	< 1	< 1	0.0124	0.0129	0.00825



Town of Ladysmith Arbutus DWTP

Monthly LRV and Turbidity Report

05/01/2024 - 06/01/2024

LRV Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	LL	LCL	%In	% between L and LL	% below LL	Unit
UF 1	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 2	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#
UF 3	LRV		5.0	0.0	32	4.25	4.5	100 %	0 %	0 %	#

LRV Daily Values

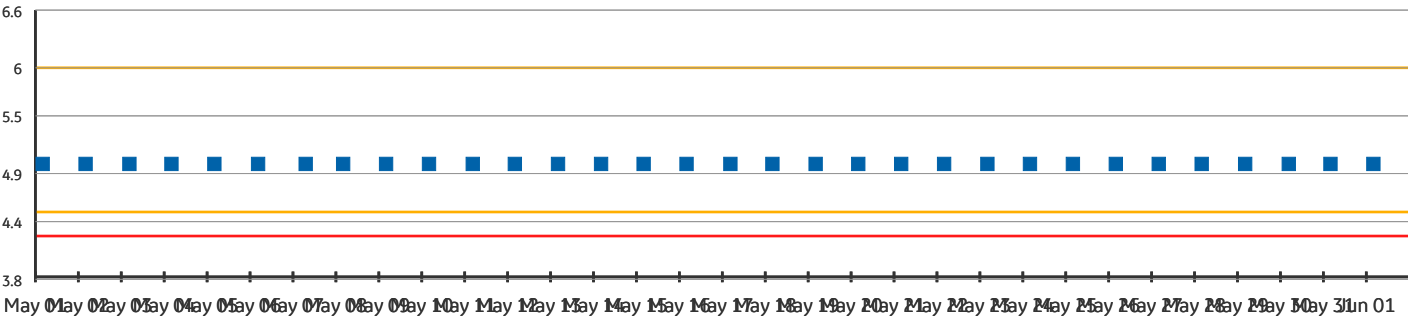
Asset	Parameter	May 01	May 02	May 03	May 04	May 05	May 06	May 07	May 08	May 09	May 10
UF 1	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	LRV	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

Asset	May 11	May 12	May 13	May 14	May 15	May 16	May 17	May 18	May 19	May 20	May 21	May 22	May 23
UF 1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

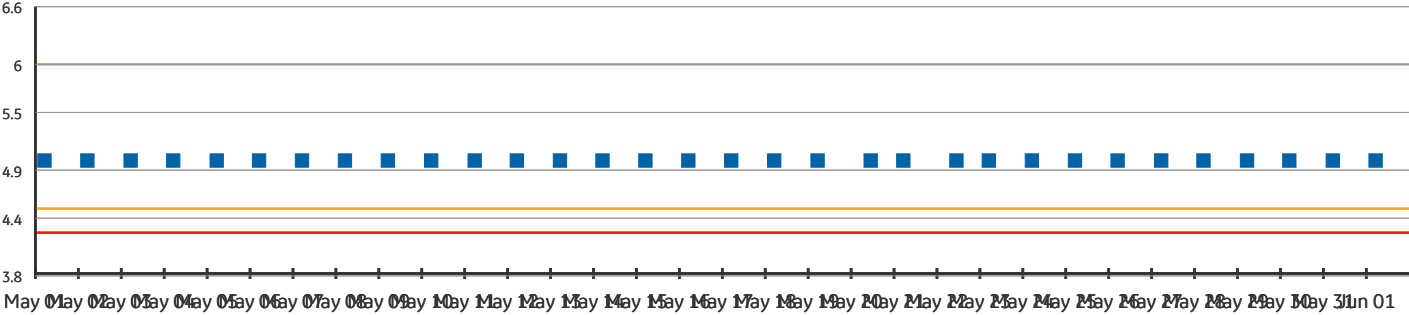
Asset	May 24	May 25	May 26	May 27	May 28	May 29	May 30	May 31	Jun 01
UF 1	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 2	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
UF 3	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0

LRV Raw Data

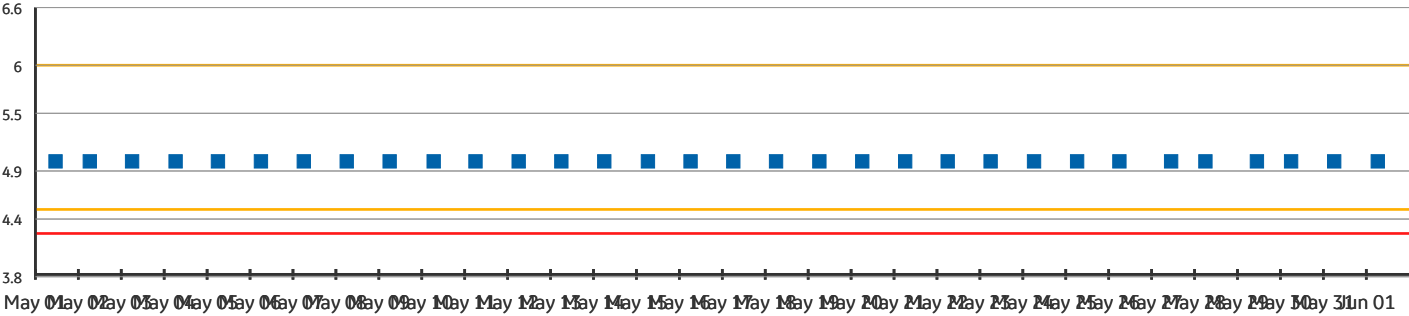
UF 1 - LRV (#)



UF 2 - LRV (#)



UF 3 - LRV (#)



Turbidity Monthly Average

Asset	Parameter	Health	Avg	Std. Dev	Points	UCL	HH	%In	% between HH and HH	% above HH	Unit
UF 1	PermeateTurbidity		0.014	0.0	45208	--	--	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidity		0.012	0.0	45208	--	--	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidity		0.014	0.0	45208	--	--	100 %	0 %	0 %	NTU
UF 1	PermeateTurbidityAfterBP		0.014	0.0	485	0.1	0.3	100 %	0 %	0 %	NTU
UF 2	PermeateTurbidityAfterBP		0.012	0.0	515	0.1	0.3	100 %	0 %	0 %	NTU
UF 3	PermeateTurbidityAfterBP		0.014	0.0	526	0.1	0.3	100 %	0 %	0 %	NTU

Turbidity Daily Averages

Asset	Parameter	May 01	May 02	May 03	May 04	May 05	May 06	May 07	May 08	May 09	May 10
UF 1	PermeateTurbidity	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.014	0.014
UF 2	PermeateTurbidity	0.011	0.012	0.012	0.011	0.011	0.011	0.012	0.012	0.011	0.012
UF 3	PermeateTurbidity	0.012	0.011	0.012	0.012	0.012	0.013	0.013	0.013	0.013	0.013
UF 1	PermeateTurbidityAfterBP	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.014	0.014
UF 2	PermeateTurbidityAfterBP	0.011	0.012	0.012	0.011	0.011	0.011	0.012	0.011	0.011	0.012
UF 3	PermeateTurbidityAfterBP	0.012	0.012	0.013	0.012	0.013	0.013	0.013	0.013	0.013	0.013

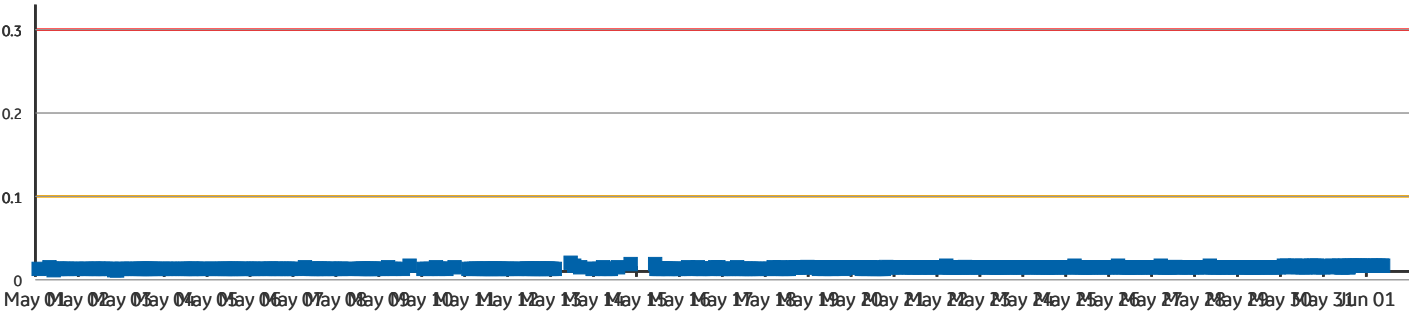
Asset	May 11	May 12	May 13	May 14	May 15	May 16	May 17	May 18	May 19	May 20	May 21	May 22	May 23
UF 1	0.013	0.013	0.015	0.014	0.015	0.013	0.013	0.013	0.013	0.013	0.015	0.015	0.015
UF 2	0.013	0.011	0.011	0.012	0.012	0.012	0.012	0.013	0.013	0.014	0.013	0.014	0.013
UF 3	0.013	0.016	0.014	0.013	0.014	0.013	0.014	0.015	0.015	0.015	0.015	0.015	0.015
UF 1	0.013	0.013	0.015	0.014	0.014	0.014	0.013	0.014	0.014	0.014	0.015	0.015	0.015
UF 2	0.012	0.012	0.012	0.012	0.012	0.012	0.013	0.013	0.013	0.014	0.013	0.014	0.013

Asset	May 11	May 12	May 13	May 14	May 15	May 16	May 17	May 18	May 19	May 20	May 21	May 22	May 23
UF 3	0.014	0.015	0.013	0.014	0.014	0.014	0.014	0.015	0.015	0.015	0.015	0.015	0.015

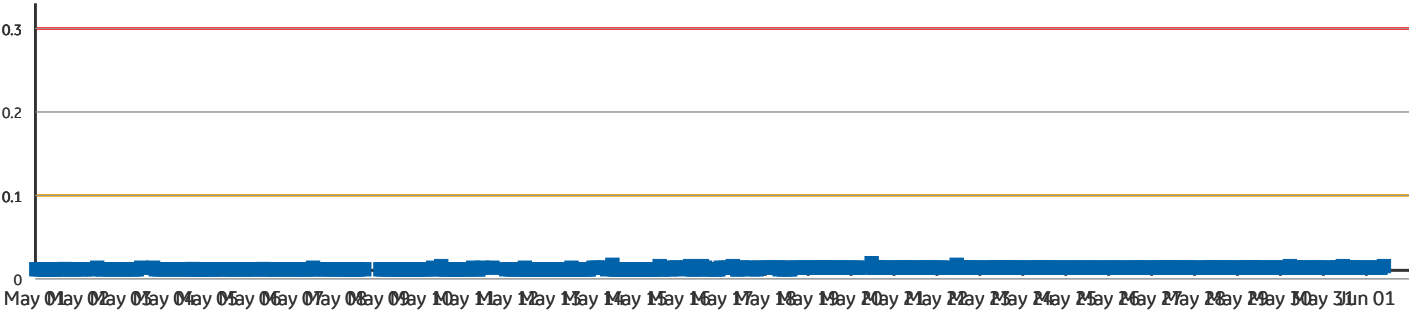
Asset	May 24	May 25	May 26	May 27	May 28	May 29	May 30	May 31	Jun 01
UF 1	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.016	0.017
UF 2	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.013
UF 3	0.015	0.015	0.015	0.016	0.015	0.015	0.015	0.015	0.014
UF 1	0.015	0.015	0.015	0.015	0.015	0.015	0.016	0.016	0.017
UF 2	0.013	0.013	0.013	0.013	0.013	0.013	0.013	0.014	0.014
UF 3	0.015	0.015	0.015	0.016	0.015	0.016	0.015	0.014	0.015

Turbidity Raw Data

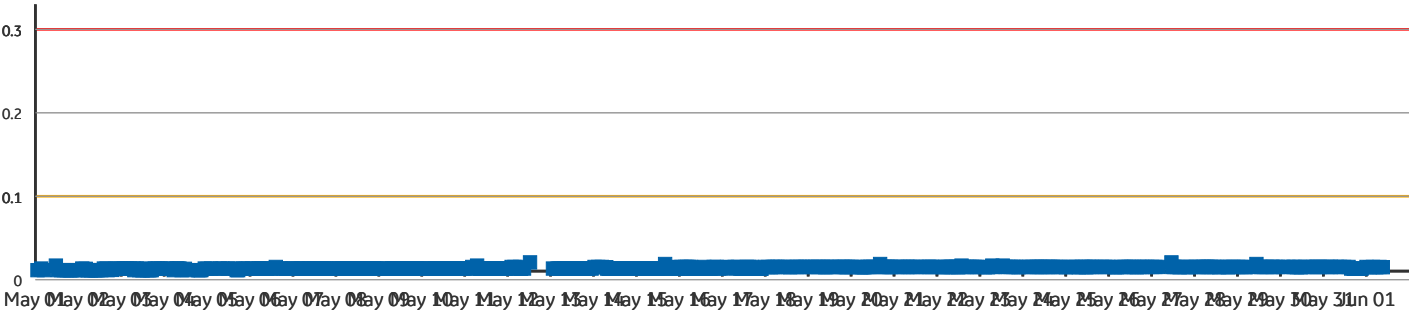
UF 1 - PermeateTurbidityAfterBP (NTU)



UF 2 - PermeateTurbidityAfterBP (NTU)



UF 3 - PermeateTurbidityAfterBP (NTU)



CERTIFICATE OF ANALYSIS

Work Order	: VA24A9697	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 03-May-2024 12:33
PO	: PO #10916	Date Analysis Commenced	: 03-May-2024
C-O-C number	: ----	Issue Date	: 08-May-2024 13:28
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	Position	Laboratory Department
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia
Monica Ko	Lab Assistant	Inorganics, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia



General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	Description
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

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>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.



Analytical Results

Sub-Matrix: Water					Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
(Matrix: Water)										
Client sampling date / time					02-May-2024 10:30	02-May-2024 10:30	02-May-2024 10:30	02-May-2024 10:30	----	
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24A9697-001	VA24A9697-002	VA24A9697-003	VA24A9697-004	-----	
					Result	Result	Result	Result	----	
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	11.0	----	
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----	
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	54.8	----	
pH	----	E108/VA	0.10	pH units	----	----	----	7.30	----	
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----	
Organic / Inorganic Carbon										
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	3.03	1.06	1.01	----	----	
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	3.17	1.40	1.04	----	----	
Microbiological Tests										
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	1	----	
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24A9697	Page	: 1 of 7
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 03-May-2024 12:33
PO	: PO #10916	Issue Date	: 08-May-2024 13:28
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.

CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.

DQO: Data Quality Objective.

LOR: Limit of Reporting (detection limit).

RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "----" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B0047	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 08-May-2024 10:20
PO	: 10880	Date Analysis Commenced	: 08-May-2024
C-O-C number	: ----	Issue Date	: 14-May-2024 14:09
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

<i>Signatories</i>	<i>Position</i>	<i>Laboratory Department</i>
Courtney Cox	Analyst- General	Inorganics, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia



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Qualifiers

Qualifier	Description
HTP	Sample preparation or preservation hold time was exceeded.
SFP	Sample was filtered and preserved at the laboratory.



Analytical Results

Sub-Matrix: Water					Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
(Matrix: Water)										
Client sampling date / time						07-May-2024 10:00	07-May-2024 10:00	07-May-2024 10:00	07-May-2024 10:00	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B0047-001	VA24B0047-002	VA24B0047-003	VA24B0047-004	-----	
					Result	Result	Result	Result	----	
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	12.2	----	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	55.5	----	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.35	----	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----	----
Organic / Inorganic Carbon										
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	3.12 ^{SFP}	1.19 ^{SFP}	1.30 ^{HTP, SFP}	----	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.93	1.44	1.12	----	----	----
Microbiological Tests										
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	----

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QUALITY CONTROL INTERPRETIVE REPORT

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Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 08-May-2024 10:20
PO	: 10880	Issue Date	: 14-May-2024 14:13
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
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No. of samples analysed	: 4		

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RPD: Relative Percent Difference.

Workorder Comments

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Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B0935	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Monthly Sampling	Date Samples Received	: 16-May-2024 12:35
PO	: 10880	Date Analysis Commenced	: 17-May-2024
C-O-C number	: ----	Issue Date	: 24-May-2024 15:49
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results
- Surrogate Control Limits

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

<i>Signatories</i>	<i>Position</i>	<i>Laboratory Department</i>
Janice Leung	Supervisor - Organics Instrumentation	Organics, Burnaby, British Columbia
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Metals, Burnaby, British Columbia
Sanja Risticvic	Department Manager - LCMS	LCMS, Waterloo, Ontario



General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	Description
µg/L	micrograms per litre
mg/L	milligrams per litre

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

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Analytical Results

Sub-Matrix: Water					Client sample ID	Treated Water (post reservoir)	----	----	----	----
(Matrix: Water)										
					Client sampling date / time	15-May-2024 10:30	----	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B0935-001	-----	-----	-----	-----	-----
					Result	----	----	----	----	----
Total Metals										
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0124	----	----	----	----	----
Volatile Organic Compounds [THMs]										
Bromodichloromethane	75-27-4	E611B/VA	1.0	µg/L	1.1	----	----	----	----	----
Bromoform	75-25-2	E611B/VA	1.0	µg/L	<1.0	----	----	----	----	----
Chloroform	67-66-3	E611B/VA	1.0	µg/L	11.8	----	----	----	----	----
Dibromochloromethane	124-48-1	E611B/VA	1.0	µg/L	<1.0	----	----	----	----	----
Trihalomethanes [THMs], total	----	E611B/VA	2.0	µg/L	12.9	----	----	----	----	----
Volatile Organic Compounds [THMs] Surrogates										
Bromofluorobenzene, 4-	460-00-4	E611B/VA	1.0	%	85.8	----	----	----	----	----
Difluorobenzene, 1,4-	540-36-3	E611B/VA	1.0	%	103	----	----	----	----	----
Haloacetic Acids										
Bromochloroacetic acid	5589-96-8	E750/WT	1.00	µg/L	<1.00	----	----	----	----	----
Dibromoacetic acid	631-64-1	E750/WT	1.00	µg/L	<1.00	----	----	----	----	----
Dichloroacetic acid	79-43-6	E750/WT	1.00	µg/L	4.64	----	----	----	----	----
Monobromoacetic acid	79-08-3	E750/WT	1.00	µg/L	<1.00	----	----	----	----	----
Monochloroacetic acid	79-11-8	E750/WT	1.00	µg/L	<1.00	----	----	----	----	----
Trichloroacetic acid	76-03-9	E750/WT	1.00	µg/L	3.61	----	----	----	----	----
Haloacetic acids, total [HAA5]	n/a	E750/WT	5.00	µg/L	8.25	----	----	----	----	----

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B0935	Page	: 1 of 5
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Monthly Sampling	Date Samples Received	: 16-May-2024 12:35
PO	: 10880	Issue Date	: 24-May-2024 15:49
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 1		
No. of samples analysed	: 1		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.

CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.

DQO: Data Quality Objective.

LOR: Limit of Reporting (detection limit).

RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "----" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Matrix Spike Duplicate (MSD) outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- No Analysis Holding Time Outliers exist.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B0931	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 16-May-2024 12:35
PO	: 10880	Date Analysis Commenced	: 16-May-2024
C-O-C number	: ----	Issue Date	: 23-May-2024 10:10
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

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- General Comments
- Analytical Results

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Signatories

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Signatories	Position	Laboratory Department
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia



General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	Description
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

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Analytical Results

Sub-Matrix: Water (Matrix: Water)				Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					15-May-2024 10:30	15-May-2024 10:30	15-May-2024 10:30	15-May-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B0931-001	VA24B0931-002	VA24B0931-003	VA24B0931-004	-----
					Result	Result	Result	Result	----
Physical Tests									
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	11.6	----
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	55.5	----
pH	----	E108/VA	0.10	pH units	----	----	----	7.29	----
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----
Organic / Inorganic Carbon									
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.65	0.71	0.98	----	----
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.63	1.10	0.99	----	----
Microbiological Tests									
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----

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Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B0931	Page	: 1 of 7
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 16-May-2024 12:35
PO	: 10880	Issue Date	: 23-May-2024 10:10
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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- No Method Blank value outliers occur.
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- No Laboratory Control Sample (LCS) outliers occur
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- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B1521	Page	: 1 of 6
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 23-May-2024 11:10
PO	: 10880	Date Analysis Commenced	: 23-May-2024
C-O-C number	: ----	Issue Date	: 30-May-2024 16:05
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 2		
No. of samples analysed	: 2		

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- Surrogate Control Limits

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Signatories

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Signatories	Position	Laboratory Department
Brianna Allen	Production/Validation Manager	Microbiology, Burnaby, British Columbia
Ghazaleh Khanmirzaei	Analyst	Metals, Burnaby, British Columbia
Harpreet Chawla	Team Leader - Inorganics	Inorganics, Calgary, Alberta
Janice Leung	Supervisor - Organics Instrumentation	Organics, Burnaby, British Columbia
Kim Jensen	Department Manager - Metals	Metals, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia



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Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
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Unit	Description
% T/cm	% transmittance per centimetre
µg/L	micrograms per litre
AU/cm	absorbance units per centimetre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

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Analytical Results

Sub-Matrix: Water					Client sample ID	Stocking Lake	Holland Lake	----	----	----
(Matrix: Water)										
					Client sampling date / time	22-May-2024 08:07	22-May-2024 10:15	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B1521-001	VA24B1521-002	-----	-----	-----	
					Result	Result	----	----	----	
Physical Tests										
Absorbance, UV (@ 254nm)	----	E404/VA	0.0050	AU/cm	0.0690	0.114	----	----	----	
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	9.5	3.6	----	----	----	
Colour, true	----	E329/VA	5.0	CU	8.0	20.8	----	----	----	
Hardness (as CaCO3), from total Ca/Mg	----	EC100A/VA	0.60	mg/L	10.3	4.75	----	----	----	
pH	----	E108/VA	0.10	pH units	7.27	6.77	----	----	----	
Solids, total dissolved [TDS]	----	E162/VA	10	mg/L	28	21	----	----	----	
Turbidity	----	E121/VA	0.10	NTU	0.57	0.50	----	----	----	
Transmittance, UV (@ 254nm)	----	E404/VA	1.0	% T/cm	85.3	76.9	----	----	----	
Anions and Nutrients										
Bromide	24959-67-9	E235.Br-L/VA	0.050	mg/L	<0.050	<0.050	----	----	----	
Organic / Inorganic Carbon										
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.62	3.47	----	----	----	
Carbon, total organic [TOC]	----	E355-L/CG	0.50	mg/L	3.60	3.36	----	----	----	
Microbiological Tests										
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	2	<1	----	----	----	
Coliforms, total	----	E010/VA	1	MPN/100mL	23	248	----	----	----	
Total Metals										
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0392	0.0877	----	----	----	
Antimony, total	7440-36-0	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----	
Arsenic, total	7440-38-2	E420/VA	0.00010	mg/L	0.00010	<0.00010	----	----	----	
Barium, total	7440-39-3	E420/VA	0.00010	mg/L	0.00311	0.00359	----	----	----	
Beryllium, total	7440-41-7	E420/VA	0.000100	mg/L	<0.000100	<0.000100	----	----	----	
Bismuth, total	7440-69-9	E420/VA	0.000050	mg/L	<0.000050	<0.000050	----	----	----	
Boron, total	7440-42-8	E420/VA	0.010	mg/L	0.019	<0.010	----	----	----	
Cadmium, total	7440-43-9	E420/VA	0.0000050	mg/L	0.000325	<0.0000050	----	----	----	
Calcium, total	7440-70-2	E420/VA	0.050	mg/L	3.34	1.51	----	----	----	
Cesium, total	7440-46-2	E420/VA	0.000010	mg/L	<0.000010	<0.000010	----	----	----	
Chromium, total	7440-47-3	E420/VA	0.00050	mg/L	<0.00050	<0.00050	----	----	----	
Cobalt, total	7440-48-4	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----	



Analytical Results

Sub-Matrix: Water					Client sample ID	Stocking Lake	Holland Lake	----	----	----
(Matrix: Water)										
Client sampling date / time					22-May-2024 08:07	22-May-2024 10:15	----	----	----	
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B1521-001	VA24B1521-002	-----	-----	-----	
					Result	Result	----	----	----	
Total Metals										
Copper, total	7440-50-8	E420/VA	0.00050	mg/L	0.00087	0.00074	----	----	----	
Iron, total	7439-89-6	E420/VA	0.010	mg/L	0.035	0.107	----	----	----	
Lead, total	7439-92-1	E420/VA	0.000050	mg/L	<0.000050	<0.000050	----	----	----	
Lithium, total	7439-93-2	E420/VA	0.0010	mg/L	<0.0010	<0.0010	----	----	----	
Magnesium, total	7439-95-4	E420/VA	0.0050	mg/L	0.474	0.238	----	----	----	
Manganese, total	7439-96-5	E420/VA	0.00010	mg/L	0.00367	0.0185	----	----	----	
Mercury, total	7439-97-6	E508/VA	0.0000050	mg/L	<0.0000050	<0.0000050	----	----	----	
Molybdenum, total	7439-98-7	E420/VA	0.000050	mg/L	0.000190	<0.000050	----	----	----	
Nickel, total	7440-02-0	E420/VA	0.00050	mg/L	<0.00050	<0.00050	----	----	----	
Phosphorus, total	7723-14-0	E420/VA	0.050	mg/L	<0.050	<0.050	----	----	----	
Potassium, total	7440-09-7	E420/VA	0.050	mg/L	0.257	0.116	----	----	----	
Rubidium, total	7440-17-7	E420/VA	0.00020	mg/L	0.00062	0.00035	----	----	----	
Selenium, total	7782-49-2	E420/VA	0.000050	mg/L	<0.000050	<0.000050	----	----	----	
Silicon, total	7440-21-3	E420/VA	0.10	mg/L	1.74	1.18	----	----	----	
Silver, total	7440-22-4	E420/VA	0.000010	mg/L	<0.000010	<0.000010	----	----	----	
Sodium, total	7440-23-5	E420/VA	0.050	mg/L	1.12	0.684	----	----	----	
Strontium, total	7440-24-6	E420/VA	0.00020	mg/L	0.00982	0.00612	----	----	----	
Sulfur, total	7704-34-9	E420/VA	0.50	mg/L	<0.50	<0.50	----	----	----	
Tellurium, total	13494-80-9	E420/VA	0.00020	mg/L	<0.00020	<0.00020	----	----	----	
Thallium, total	7440-28-0	E420/VA	0.000010	mg/L	<0.000010	<0.000010	----	----	----	
Thorium, total	7440-29-1	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----	
Tin, total	7440-31-5	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----	
Titanium, total	7440-32-6	E420/VA	0.00030	mg/L	0.00040	0.00064	----	----	----	
Tungsten, total	7440-33-7	E420/VA	0.00010	mg/L	<0.00010	<0.00010	----	----	----	
Uranium, total	7440-61-1	E420/VA	0.000010	mg/L	0.000021	<0.000010	----	----	----	
Vanadium, total	7440-62-2	E420/VA	0.00050	mg/L	<0.00050	<0.00050	----	----	----	
Zinc, total	7440-66-6	E420/VA	0.0030	mg/L	<0.0030	<0.0030	----	----	----	
Zirconium, total	7440-67-7	E420/VA	0.00020	mg/L	<0.00020	<0.00020	----	----	----	
Hydrocarbons										
EPH (C10-C19)	----	E601A/VA	250	µg/L	<250	<250	----	----	----	



Analytical Results

Sub-Matrix: Water					Client sample ID	Stocking Lake	Holland Lake	----	----	----
(Matrix: Water)					Client sampling date / time	22-May-2024 08:07	22-May-2024 10:15	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B1521-001	VA24B1521-002	-----	-----	-----	
					Result	Result	----	----	----	
Hydrocarbons										
EPH (C10-C32)	----	E601A/VA	400	µg/L	<400	<400	----	----	----	
EPH (C19-C32)	----	E601A/VA	250	µg/L	<250	<250	----	----	----	
TEH (C10-C30), BC	----	E601A/VA	250	µg/L	<250	<250	----	----	----	
Hydrocarbons Surrogates										
Bromobenzotrifluoride, 2- (EPH surrogate)	392-83-6	E601A/VA	1.0	%	77.3	74.9	----	----	----	
Polycyclic Aromatic Hydrocarbons										
Acenaphthene	83-32-9	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Acenaphthylene	208-96-8	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Acridine	260-94-6	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Anthracene	120-12-7	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Benz(a)anthracene	56-55-3	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Benzo(a)pyrene	50-32-8	E641A/VA	0.0050	µg/L	<0.0050	<0.0050	----	----	----	
Benzo(b+j)fluoranthene	n/a	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Benzo(b+j+k)fluoranthene	n/a	E641A/VA	0.015	µg/L	<0.015	<0.015	----	----	----	
Benzo(g,h,i)perylene	191-24-2	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Benzo(k)fluoranthene	207-08-9	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Chrysene	218-01-9	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Dibenz(a,h)anthracene	53-70-3	E641A/VA	0.0050	µg/L	<0.0050	<0.0050	----	----	----	
Fluoranthene	206-44-0	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Fluorene	86-73-7	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Indeno(1,2,3-c,d)pyrene	193-39-5	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Methylnaphthalene, 1-	90-12-0	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Methylnaphthalene, 2-	91-57-6	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Naphthalene	91-20-3	E641A/VA	0.050	µg/L	<0.050	<0.050	----	----	----	
Phenanthrene	85-01-8	E641A/VA	0.020	µg/L	<0.020	<0.020	----	----	----	
Pyrene	129-00-0	E641A/VA	0.010	µg/L	<0.010	<0.010	----	----	----	
Quinoline	91-22-5	E641A/VA	0.050	µg/L	<0.050	<0.050	----	----	----	
Polycyclic Aromatic Hydrocarbons Surrogates										
Chrysene-d12	1719-03-5	E641A/VA	0.1	%	97.6	109	----	----	----	
Naphthalene-d8	1146-65-2	E641A/VA	0.1	%	94.0	98.2	----	----	----	



Analytical Results

Sub-Matrix: Water					Client sample ID	Stocking Lake	Holland Lake	----	----	----
(Matrix: Water)										
					Client sampling date / time	22-May-2024 08:07	22-May-2024 10:15	----	----	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B1521-001	VA24B1521-002	-----	-----	-----	
					Result	Result	----	----	----	
Polycyclic Aromatic Hydrocarbons Surrogates										
Phenanthrene-d10	1517-22-2	E641A/VA	0.1	%	104	113	----	----	----	

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B1521	Page	: 1 of 11
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Weekly Sampling	Date Samples Received	: 23-May-2024 11:10
PO	: 10880	Issue Date	: 30-May-2024 16:05
C-O-C number	: ----		
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 2		
No. of samples analysed	: 2		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.

CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.

DQO: Data Quality Objective.

LOR: Limit of Reporting (detection limit).

RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "----" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B1513	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 23-May-2024 11:10
PO	: PO #10916	Date Analysis Commenced	: 23-May-2024
C-O-C number	: ----	Issue Date	: 29-May-2024 12:51
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	Position	Laboratory Department
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Microbiology, Burnaby, British Columbia



General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

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Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

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Unit	Description
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Sample Treated Water (post reservoir): HPC Exceeded Recommended Holding Time prior to receipt at the lab for Microbiology analysis. Testing will proceed. No BC/Yukon Drinking Water Declaration form was received.



Analytical Results

Sub-Matrix: Water					Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
(Matrix: Water)										
					Client sampling date / time	22-May-2024 10:30	22-May-2024 10:30	22-May-2024 10:30	22-May-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B1513-001	VA24B1513-002	VA24B1513-003	VA24B1513-004	-----	
					Result	Result	Result	Result	----	
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	----	----	----	11.8	----	
Colour, true	----	E329/VA	5.0	CU	----	----	----	<5.0	----	
Conductivity	----	E100/VA	2.0	µS/cm	----	----	----	57.9	----	
pH	----	E108/VA	0.10	pH units	----	----	----	7.38	----	
Turbidity	----	E121/VA	0.10	NTU	----	----	----	<0.10	----	
Organic / Inorganic Carbon										
Carbon, dissolved organic [DOC]	----	E358-L/VA	0.50	mg/L	2.85	0.91	0.91	----	----	
Carbon, total organic [TOC]	----	E355-L/VA	0.50	mg/L	2.54	1.26	0.92	----	----	
Microbiological Tests										
Heterotrophic plate count [HPC]	----	E020/VA	1	CFU/mL	----	----	----	<1	----	
Coliforms, Escherichia coli [E. coli]	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	
Coliforms, total	----	E010/VA	1	MPN/100mL	----	----	----	<1	----	

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Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B1513	Page	: 1 of 7
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 23-May-2024 11:10
PO	: PO #10916	Issue Date	: 29-May-2024 12:50
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B2518	Page	: 1 of 4
Amendment	: 1		
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Quarterly DT Sampling	Date Samples Received	: 31-May-2024 12:00
PO	: 10880	Date Analysis Commenced	: 04-Jun-2024
C-O-C number	: ----	Issue Date	: 11-Jun-2024 10:44
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	Position	Laboratory Department
Angela Ren	Team Leader - Metals	Metals, Burnaby, British Columbia
Miles Gropen	Department Manager - Inorganics	Inorganics, Burnaby, British Columbia
Tracy Harley	Supervisor - Water Quality Instrumentation	Inorganics, Burnaby, British Columbia



General Comments

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Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	Description
mg/L	milligrams per litre
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Amendment (11/06/2024): This report has been amended and re-released to allow the reporting of additional analytical data.

MB hit for Alkalinity but all the samples were re-analysed



Analytical Results

Sub-Matrix: Water					Client sample ID	FJCC	Town Hall	Fire Department	RCMP	----
(Matrix: Water)										
Client sampling date / time						30-May-2024 10:30	30-May-2024 10:30	30-May-2024 10:30	30-May-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B2518-001	VA24B2518-002	VA24B2518-003	VA24B2518-004	-----	
					Result	Result	Result	Result	----	
Physical Tests										
Alkalinity, total (as CaCO3)	----	E290/VA	1.0	mg/L	13.0	13.7	12.6	13.7	----	
pH	----	E108/VA	0.10	pH units	7.35	7.36	7.37	7.43	----	
Solids, total dissolved [TDS]	----	E162/VA	10	mg/L	46	39	40	44	----	
Total Metals										
Aluminum, total	7429-90-5	E420/VA	0.0030	mg/L	0.0102	0.0101	0.0093	0.0460	----	
Antimony, total	7440-36-0	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Arsenic, total	7440-38-2	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Barium, total	7440-39-3	E420/VA	0.00010	mg/L	0.00330	0.00346	0.00349	0.00310	----	
Beryllium, total	7440-41-7	E420/VA	0.000020	mg/L	<0.000020	<0.000020	<0.000020	<0.000020	----	
Bismuth, total	7440-69-9	E420/VA	0.000050	mg/L	<0.000050	<0.000050	<0.000050	<0.000050	----	
Boron, total	7440-42-8	E420/VA	0.010	mg/L	0.020	0.021	0.021	0.023	----	
Cadmium, total	7440-43-9	E420/VA	0.0000050	mg/L	<0.0000050	<0.0000050	<0.0000050	<0.0000050	----	
Calcium, total	7440-70-2	E420/VA	0.050	mg/L	2.31	2.24	2.29	2.30	----	
Cesium, total	7440-46-2	E420/VA	0.000010	mg/L	<0.000010	<0.000010	<0.000010	<0.000010	----	
Chromium, total	7440-47-3	E420/VA	0.00050	mg/L	<0.00050	<0.00050	<0.00050	<0.00050	----	
Cobalt, total	7440-48-4	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Copper, total	7440-50-8	E420/VA	0.00050	mg/L	0.0392	0.0787	0.343	0.241	----	
Iron, total	7439-89-6	E420/VA	0.010	mg/L	0.013	<0.010	<0.010	0.284	----	
Lead, total	7439-92-1	E420/VA	0.000050	mg/L	0.000614	0.000302	0.00203	0.00103	----	
Lithium, total	7439-93-2	E420/VA	0.0010	mg/L	<0.0010	<0.0010	<0.0010	<0.0010	----	
Magnesium, total	7439-95-4	E420/VA	0.0050	mg/L	0.352	0.390	0.372	0.564	----	
Manganese, total	7439-96-5	E420/VA	0.00010	mg/L	0.00016	<0.00010	0.00017	0.00215	----	
Molybdenum, total	7439-98-7	E420/VA	0.000050	mg/L	0.000062	0.000068	0.000087	0.000075	----	
Nickel, total	7440-02-0	E420/VA	0.00050	mg/L	<0.00050	<0.00050	0.00053	<0.00050	----	
Phosphorus, total	7723-14-0	E420/VA	0.050	mg/L	<0.050	<0.050	<0.050	<0.050	----	
Potassium, total	7440-09-7	E420/VA	0.050	mg/L	0.138	0.137	0.135	0.142	----	
Rubidium, total	7440-17-7	E420/VA	0.00020	mg/L	0.00035	0.00035	0.00036	0.00033	----	
Selenium, total	7782-49-2	E420/VA	0.000050	mg/L	<0.000050	<0.000050	<0.000050	<0.000050	----	
Silicon, total	7440-21-3	E420/VA	0.10	mg/L	2.15	2.23	2.09	2.13	----	



Analytical Results

Sub-Matrix: Water					Client sample ID	FJCC	Town Hall	Fire Department	RCMP	----
(Matrix: Water)										
					Client sampling date / time	30-May-2024 10:30	30-May-2024 10:30	30-May-2024 10:30	30-May-2024 10:30	----
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B2518-001	VA24B2518-002	VA24B2518-003	VA24B2518-004	-----	
					Result	Result	Result	Result	----	
Total Metals										
Silver, total	7440-22-4	E420/VA	0.000010	mg/L	<0.000010	<0.000010	0.000025	<0.000010	----	
Sodium, total	7440-23-5	E420/VA	0.050	mg/L	8.23	8.62	8.43	8.51	----	
Strontium, total	7440-24-6	E420/VA	0.00020	mg/L	0.0109	0.0116	0.0114	0.0112	----	
Sulfur, total	7704-34-9	E420/VA	0.50	mg/L	0.63	0.68	0.65	0.68	----	
Tellurium, total	13494-80-9	E420/VA	0.00020	mg/L	<0.00020	<0.00020	<0.00020	<0.00020	----	
Thallium, total	7440-28-0	E420/VA	0.000010	mg/L	<0.000010	<0.000010	<0.000010	<0.000010	----	
Thorium, total	7440-29-1	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Tin, total	7440-31-5	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Titanium, total	7440-32-6	E420/VA	0.00030	mg/L	<0.00030	<0.00030	<0.00030	<0.00030	----	
Tungsten, total	7440-33-7	E420/VA	0.00010	mg/L	<0.00010	<0.00010	<0.00010	<0.00010	----	
Uranium, total	7440-61-1	E420/VA	0.000010	mg/L	<0.000010	<0.000010	<0.000010	<0.000010	----	
Vanadium, total	7440-62-2	E420/VA	0.00050	mg/L	<0.00050	<0.00050	<0.00050	<0.00050	----	
Zinc, total	7440-66-6	E420/VA	0.0030	mg/L	0.0042	0.0044	0.0085	0.0134	----	
Zirconium, total	7440-67-7	E420/VA	0.00020	mg/L	<0.00020	<0.00020	<0.00020	<0.00020	----	

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Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B2518	Page	: 1 of 8
Amendment	: 1		
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment - Quarterly DT Sampling	Date Samples Received	: 31-May-2024 12:00
PO	: 10880	Issue Date	: 11-Jun-2024 10:46
C-O-C number	: ----		
Sampler	: ----		
Site	: ----		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

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CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.

DQO: Data Quality Objective.

LOR: Limit of Reporting (detection limit).

RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "----" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- Method Blank value outliers occur - please see following pages for full details.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.

CERTIFICATE OF ANALYSIS

Work Order	: VA24B2424	Page	: 1 of 3
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby BC Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 31-May-2024 12:00
PO	: PO #10916	Date Analysis Commenced	: 31-May-2024
C-O-C number	: ----	Issue Date	: 06-Jun-2024 10:56
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report supersedes any previous report(s) with this reference. Results apply to the sample(s) as submitted. This document shall not be reproduced, except in full.

This Certificate of Analysis contains the following information:

- General Comments
- Analytical Results

Additional information pertinent to this report will be found in the following separate attachments: Quality Control Report, QC Interpretive report to assist with Quality Review and Sample Receipt Notification (SRN).

Signatories

This document has been electronically signed by the authorized signatories below. Electronic signing is conducted in accordance with US FDA 21 CFR Part 11.

Signatories	Position	Laboratory Department
Kate Dimitrova	Supervisor - Inorganic	Inorganics, Burnaby, British Columbia
Kevin Duarte	Supervisor - Metals ICP Instrumentation	Microbiology, Burnaby, British Columbia
Monica Ko	Lab Assistant	Microbiology, Burnaby, British Columbia



General Comments

The analytical methods used by ALS are developed using internationally recognized reference methods (where available), such as those published by US EPA, APHA Standard Methods, ASTM, ISO, Environment Canada, BC MOE, and Ontario MOE. Refer to the ALS Quality Control Interpretive report (QCI) for applicable references and methodology summaries. Reference methods may incorporate modifications to improve performance.

Where a reported less than (<) result is higher than the LOR, this may be due to primary sample extract/digestate dilution and/or insufficient sample for analysis.

Where the LOR of a reported result differs from standard LOR, this may be due to high moisture content, insufficient sample (reduced weight employed) or matrix interference.

Please refer to Quality Control Interpretive report (QCI) for information regarding Holding Time compliance.

Key : CAS Number: Chemical Abstracts Services number is a unique identifier assigned to discrete substances
LOR: Limit of Reporting (detection limit).

Unit	Description
µS/cm	microsiemens per centimetre
CFU/mL	colony forming units per millilitre
CU	colour units (1 cu = 1 mg/l pt)
mg/L	milligrams per litre
MPN/100mL	most probable number per hundred millilitres
NTU	nephelometric turbidity units
pH units	pH units

<: less than.

>: greater than.

Surrogate: An analyte that is similar in behavior to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.

Test results reported relate only to the samples as received by the laboratory.

UNLESS OTHERWISE STATED on SRN or QCI Report, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.

Workorder Comments

Sample Treated Water (post reservoir): HPC Exceeded Recommended Holding Time prior to receipt at the lab for Microbiology analysis. Testing will proceed.



Analytical Results

Sub-Matrix: Water (Matrix: Water)					Client sample ID	Raw Water	DAF Effluent	UF Effluent	Treated Water (post reservoir)	----
Client sampling date / time					30-May-2024 09:00	30-May-2024 09:00	30-May-2024 09:00	30-May-2024 09:00	----	
Analyte	CAS Number	Method/Lab	LOR	Unit	VA24B2424-001	VA24B2424-002	VA24B2424-003	VA24B2424-004	-----	
					Result	Result	Result	Result	----	
Physical Tests										
Alkalinity, total (as CaCO3)	---	E290/VA	1.0	mg/L	----	----	----	12.3	----	
Colour, true	---	E329/VA	5.0	CU	----	----	----	<5.0	----	
Conductivity	---	E100/VA	2.0	µS/cm	----	----	----	56.5	----	
pH	---	E108/VA	0.10	pH units	----	----	----	7.28	----	
Turbidity	---	E121/VA	0.10	NTU	----	----	----	<0.10	----	
Organic / Inorganic Carbon										
Carbon, dissolved organic [DOC]	---	E358-L/VA	0.50	mg/L	2.35	0.72	0.82	----	----	
Carbon, total organic [TOC]	---	E355-L/VA	0.50	mg/L	2.76	1.83	0.96	----	----	
Microbiological Tests										
Heterotrophic plate count [HPC]	---	E020/VA	1	CFU/mL	----	----	----	<1	----	
Coliforms, Escherichia coli [E. coli]	---	E010/VA	1	MPN/100mL	----	----	----	<1	----	
Coliforms, total	---	E010/VA	1	MPN/100mL	----	----	----	<1	----	

Please refer to the General Comments section for an explanation of any result qualifiers detected.

Please refer to the Accreditation section for an explanation of analyte accreditations.

QUALITY CONTROL INTERPRETIVE REPORT

Work Order	: VA24B2424	Page	: 1 of 7
Client	: Town of Ladysmith	Laboratory	: ALS Environmental - Vancouver
Contact	: Shawn Baker	Account Manager	: Thomas Chang
Address	: 410 Esplanade PO Box 220 Ladysmith BC Canada V9G 1A2	Address	: 8081 Lougheed Highway Burnaby, British Columbia Canada V5A 1W9
Telephone	: ----	Telephone	: +1 604 253 4188
Project	: Arbutus Water Treatment Plant - Weekly Sampling	Date Samples Received	: 31-May-2024 12:00
PO	: PO #10916	Issue Date	: 06-Jun-2024 10:57
C-O-C number	: ----		
Sampler	: ----		
Site	: Town of Ladysmith		
Quote number	: VA22-GMSM100-001 Tender# 2022-IS-20		
No. of samples received	: 4		
No. of samples analysed	: 4		

This report is automatically generated by the ALS LIMS (Laboratory Information Management System) through evaluation of Quality Control (QC) results and other QA parameters associated with this submission, and is intended to facilitate rapid data validation by auditors or reviewers. The report highlights any exceptions and outliers to ALS Data Quality Objectives, provides holding time details and exceptions, summarizes QC sample frequencies, and lists applicable methodology references and summaries.

Key

Anonymous: Refers to samples which are not part of this work order, but which formed part of the QC process lot.

CAS Number: Chemical Abstracts Service number is a unique identifier assigned to discrete substances.

DQO: Data Quality Objective.

LOR: Limit of Reporting (detection limit).

RPD: Relative Percent Difference.

Workorder Comments

Holding times are displayed as "----" if no guidance exists from CCME, Canadian provinces, or broadly recognized international references.

Summary of Outliers

Outliers : Quality Control Samples

- No Method Blank value outliers occur.
- No Duplicate outliers occur.
- No Laboratory Control Sample (LCS) outliers occur
- No Matrix Spike outliers occur.
- No Test sample Surrogate recovery outliers exist.

Outliers: Reference Material (RM) Samples

- No Reference Material (RM) Sample outliers occur.

Outliers : Analysis Holding Time Compliance (Breaches)

- Analysis Holding Time Outliers exist - please see following pages for full details.

Outliers : Frequency of Quality Control Samples

- No Quality Control Sample Frequency Outliers occur.